

Arrhan Bhatia

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EDUCATION

University of Illinois Urbana Champaign

May 2026

Bachelor of Science in Computer Engineering, Minor in Hoefft Technology and Management Program

GPA: 3.90

- Robert M. Janowiak Scholarship Recipient
- James Scholar honors program and Dean's List

Coursework: Operating Systems, Data Structures, Algorithms, Machine Learning, Database Systems

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, Java, TypeScript, C#, PowerShell

Frameworks/Tools: React.js, Next.js, MongoDB, PostgreSQL, Node.js, AWS, Azure, Tensorflow, PySpark, Supabase

EXPERIENCE

Microsoft

May 2025 – August 2025

Software Engineering Intern - Azure High Performance Computing & AI

- Conceptualized and developed a secure internal tool from the ground up to automate ad-hoc script deployments in Azure, eliminating legacy processes and ensuring greater compliance with Microsoft's Safe Deployment Practices
- Designed a reusable plugin-based architecture with a shared function library, enabling engineers to quickly build and run new plugins across diverse use cases
- Owned the full product lifecycle from technical exploration and design to delivering production-ready code validated in live test environments and staged for rollout
- Presented the tool to senior leadership, communicating its technical architecture and business value, securing approval for Azure-wide adoption, with availability to engineers across the organization

National Center of Supercomputing Applications

May 2024 – March 2025

Software Engineering Intern

- Developed ML models to identify key predictors and forecast future settlement patterns of Syrian refugee distribution in Lebanon with the findings now being prepared for publication in two academic papers
- Collaborated with UN advisors to inform humanitarian planning and guide allocation of over \$53 million in aid
- Automated image processing pipeline with PaddleOCR to extract refugee counts across 1,600+ districts per map, reducing manual data entry workload by 10+ hours per map

Realty Income

June 2024 – August 2024

Machine Learning Intern - Predictive Analytics

- Developed a ML model predicting 84% of big-box retail store closures, improving risk assessment for investments
- Implemented a fuzzy string matching algorithm to unify tenant data across datasets and improve data integration

University of Illinois Department of Political Science

January 2024 – May 2024

Software Engineer

- Built a real-time sentiment analysis dashboard processing 5,000+ social media posts daily and creating visualizations; tool was adopted by 50+ researchers to explore political discourse trends
- Engineered a data analysis pipeline using Pandas and Python to add quantitative metrics to mentor's dissertation

Advanced Micro Devices

January 2024 – May 2024

Software Engineer

- Worked with engineers to build an ML-based malware detection tool using hardware performance counters
- Built an automated data collection pipeline (Bash, Python) integrated with AMD's proprietary tools, streamlining performance counter collection across diverse workloads as part of University of Illinois Disruption Lab

PROJECTS

Credit Card Recommendation Platform

September 2025

- Architected a full-stack credit card recommendation engine using Next.js and Supabase to provide users with personalized suggestions saving users 5+ hours on credit card research
- Integrated the Plaid API to securely fetch and analyze user transaction data and enhance credit card recommendations
- Developed a custom recommendation algorithm which helps identify the top credit cards for a user based on their financial goals, spending habits and current financial standing